

## ASSESSMENT TOOL- 4

### 1) INCORRECT JOIN SYNTAX FIXED

```
mysql> USE college;
Database changed

mysql> CREATE TABLE Department(
  -> Dept_ID INT PRIMARY KEY,
  -> Dept_Name VARCHAR(20));
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE Student(
  -> Student_ID INT PRIMARY KEY,
  -> Name VARCHAR(20),
  -> Dept_ID INT,
  -> FOREIGN KEY (Dept_ID) REFERENCES
    Department(Dept_ID);
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO Department VALUES
(1,'CSE'),(2,'AI & DS');
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> INSERT INTO Student VALUES
(101,'Arun',1),(102,'Priya',2),(103,'Kiran',1);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> -- Incorrect Query (using comma join)
mysql> SELECT * FROM Student, Department
  -> WHERE Student.Dept_ID = Department.Dept_ID;
+-----+-----+-----+-----+-----+
| Student_ID | Name | Dept_ID | Dept_ID | Dept_Name |
+-----+-----+-----+-----+-----+
| 101        | Arun | 1        | 1        | CSE        |
| 102        | Priya | 2        | 2        | AI & DS    |
| 103        | Kiran | 1        | 1        | CSE        |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> -- Correct Query (using INNER JOIN)
mysql> SELECT Student.Student_ID, Student.Name,
  -> Department.Dept_Name
  -> FROM Student
  -> INNER JOIN Department
  -> ON Student.Dept_ID = Department.Dept_ID;
+-----+-----+-----+
| Student_ID | Name | Dept_Name |
+-----+-----+-----+
| 101        | Arun | CSE        |
| 102        | Priya | AI & DS    |
| 103        | Kiran | CSE        |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

## 2) CORRELATED SUBQUERY FIXED

```
mysql> USE college;
Database changed

mysql> CREATE TABLE Employee(
-> Emp_ID INT PRIMARY KEY,
-> Name VARCHAR(20),
-> Dept_ID INT,
-> Salary INT);
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Employee VALUES
(1,'Arun',1,60000),
(2,'Priya',1,55000),
(3,'Kiran',2,65000),
(4,'Meena',2,40000);
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql> -- Incorrect Sub-query (no correlation)
mysql> SELECT Name, Salary, Dept_ID
-> FROM Employee E
-> WHERE Salary > (SELECT AVG(Salary) FROM Employee);
+-----+-----+-----+
| Name   | Salary | Dept_ID |
+-----+-----+-----+
| Arun   | 60000  | 1       |
| Kiran  | 65000  | 2       |
+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> -- Corrected Sub-query (correlated)
mysql> SELECT Name, Salary, Dept_ID
-> FROM Employee E
-> WHERE Salary > (SELECT AVG(Salary)
-> FROM Employee
-> WHERE Dept_ID = E.Dept_ID);
+-----+-----+-----+
| Name   | Salary | Dept_ID |
+-----+-----+-----+
| Arun   | 60000  | 1       |
| Kiran  | 65000  | 2       |
+-----+-----+-----+
2 rows in set (0.00 sec)
```

### 3) VIEW CREATED AND QUERIED

```
mysql> USE college;
Database changed

mysql> CREATE VIEW Student_View AS
-> SELECT Student_ID, Name, Dept_ID
-> FROM Student;
Query OK, 0 rows affected (0.02 sec)

mysql> SELECT * FROM Student_View;
+-----+-----+-----+
| Student_ID | Name   | Dept_ID |
+-----+-----+-----+
| 101        | Arun   | 1        |
| 102        | Priya  | 2        |
| 103        | Kiran  | 1        |
+-----+-----+-----+
2 rows in set (0.00 sec)
```

#### 4) RELATIONAL ALGEBRA (NATURAL JOIN) DEBUGGED

```
mysql> -- Student Table
mysql> SELECT * FROM Student;
+-----+-----+-----+
| Student_ID | Name   | Dept_ID |
+-----+-----+-----+
| 101        | Arun   | 1        |
| 102        | Priya  | 2        |
| 103        | Kiran  | 1        |
+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> -- Course Table
mysql> CREATE TABLE Course(
  -> Course_ID INT PRIMARY KEY,
  -> Course_Name VARCHAR(20),
  -> Dept_ID INT);
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Course VALUES
(10, 'DBMS', 1),
(20, 'AI', 2),
(30, 'CN', 1);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Course;
+-----+-----+-----+
| Course_ID | Course_Name | Dept_ID |
+-----+-----+-----+
| 10        | DBMS        | 1        |
| 20        | AI          | 2        |
| 30        | CN          | 1        |
+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> -- Natural Join (using common attribute Dept_ID)
mysql> SELECT * FROM Student NATURAL JOIN Course;
+-----+-----+-----+-----+-----+
| Student_ID | Name   | Dept_ID | Course_ID | Course_Name |
+-----+-----+-----+-----+-----+
| 101        | Arun   | 1        | 10        | DBMS        |
| 101        | Arun   | 1        | 30        | CN          |
| 102        | Priya  | 2        | 20        | AI          |
| 103        | Kiran  | 1        | 10        | DBMS        |
| 103        | Kiran  | 1        | 30        | CN          |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

## 5) REFERENTIAL INTEGRITY ENFORCED

```
mysql> USE college;
Database changed

mysql> ALTER TABLE Employee
-> ADD CONSTRAINT fk_emp_dept
-> FOREIGN KEY (Dept_ID)
-> REFERENCES Department(Dept_ID);
Query OK, 0 rows affected (0.03 sec)

mysql> -- Trying to insert invalid Dept_ID
mysql> INSERT INTO Employee VALUES
(105,'Rohit',99,50000);
ERROR 1452 (23000): Cannot add or update a
child row: a foreign key constraint fails
('college`.`employee`, CONSTRAINT
`fk_emp_dept` FOREIGN KEY (`Dept_ID`)
REFERENCES `department` (`Dept_ID`))

mysql> -- Insert valid Department first
mysql> INSERT INTO Department VALUES (3,'IT');
Query OK, 1 row affected (0.01 sec)

mysql> -- Now insert Employee
mysql> INSERT INTO Employee VALUES
(105,'Rohit',3,50000);
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM Employee WHERE Emp_ID=105;
+-----+-----+-----+-----+
| Emp_ID | Name  | Dept_ID | Salary |
+-----+-----+-----+-----+
| 105    | Rohit | 3       | 50000  |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

## 6) INEFFICIENT SUB QUERY REWRITTEN USING JOIN

```
mysql> USE college;
Database changed

mysql> -- Sub-query version
mysql> SELECT Name
  -> FROM Employee
  -> WHERE Dept_ID IN
  -> (SELECT Dept_ID
  -> FROM Department
  -> WHERE Dept_Name = 'CSE');
+-----+
| Name |
| Arun |
| Priya |
+-----+
2 rows in set (0.00 sec)

mysql> -- Optimized JOIN version
mysql> SELECT E.Name
  -> FROM Employee E
  -> INNER JOIN Department D
  -> ON E.Dept_ID = D.Dept_ID
  -> WHERE D.Dept_Name = 'CSE';
+-----+
| Name |
| Arun |
| Priya |
+-----+
2 rows in set (0.00 sec)
```

## 7) GROUP BY QUERY FIXED

```
mysql> USE college;
Database changed

mysql> -- Incorrect GROUP BY (Name not in GROUP BY)
mysql> SELECT Dept_ID, Name, COUNT(*)
  -> FROM Employee
  -> GROUP BY Dept_ID;
ERROR 1055 (42000): Expression #2 of SELECT list
is not in GROUP BY clause and contains nonaggregated
column 'college.employee.Name' which is not functionally
dependent on columns in GROUP BY clause; this is
incompatible with sql_mode=only_full_group_by

mysql> -- Correct GROUP BY
mysql> SELECT Dept_ID, COUNT(*) AS Total_Employees
  -> FROM Employee
  -> GROUP BY Dept_ID;
+-----+-----+
| Dept_ID | Total_Employees |
+-----+-----+
| 1       | 2               |
| 2       | 2               |
| 3       | 1               |
+-----+-----+
3 rows in set (0.00 sec)
```

## 8) UPDATE QUERY FIXED

```
mysql> USE college;
Database changed
mysql> SELECT * FROM Employee;
+----+-----+-----+-----+
| Emp_ID | Name  | Dept_ID | Salary |
+----+-----+-----+-----+
| 1      | Arun  | 1       | 60000  |
| 2      | Priya | 1       | 55000  |
| 3      | Kiron | 2       | 65000  |
| 4      | Meena | 2       | 40000  |
+----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> -- Incorrect UPDATE (no WHERE clause)
mysql> UPDATE Employee SET Salary = 70000;
Query OK, 4 rows affected (0.01 sec)
Rows matched: 4  Changed: 4  Warnings: 0
mysql> SELECT * FROM Employee;
+----+-----+-----+-----+
| 1      | Arun  | 1       | 70000  |
| 2      | Priya | 1       | 70000  |
| 3      | Kiron | 2       | 70000  |
| 4      | Meena | 2       | 70000  |
+----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> -- Correct UPDATE
mysql> UPDATE Employee SET Salary = 60000 WHERE Emp_ID=1;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
mysql> SELECT * FROM Employee WHERE Emp_ID=1;
+-----+-----+-----+-----+
| Dept_ID | Name  | Employee | Salary |
+-----+-----+-----+-----+
| 1       | Arun  | 1        | 60000  |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

## 9) MULTI TABLE QUERY OPTIMIZED

```
mysql> USE college;
Database changed

mysql> -- Inefficient sub-query version
mysql> SELECT Name
  -> FROM Employee
  -> WHERE Dept_ID IN
  -> (SELECT Dept_ID FROM Department
  ->   WHERE Manager_ID IN
  ->   (SELECT Manager_ID FROM Manager
  ->     WHERE Experience > 5));
+-----+
| Name |
+-----+
| Arun |
| Priya |
| Meena |
+-----+
3 rows in set (0.00 sec)

mysql> -- Optimized JOIN version
mysql> SELECT E.Name
  -> FROM Employee E
  -> INNER JOIN Department D ON E.Dept_ID = D.Dept_ID
  -> INNER JOIN Manager M ON D.Manager_ID = M.Manager_ID
  -> WHERE M.Experience > 5;
+-----+
| Name |
+-----+
| Arun |
| Priya |
| Meena |
+-----+
3 rows in set (0.00 sec)
```

## 10) RELATIONAL CALCULUS EXPRESSION DEBUGGED

```
mysql> USE college;
Database changed

mysql> -- Employees whose department exists in Department table

mysql> SELECT E.Name
-> FROM Employee E
-> WHERE EXISTS (SELECT *
-> FROM Department D
-> WHERE D.Dept_ID = E.Dept_ID);
->
+-----+
| Name  |
+-----+
| Arun  |
| Priya |
| Kiran |
| Meena |
| Rohit |
+-----+
5 rows in set (0.00 sec)
```